

FRANCESCO LORENZI

Curriculum vitæ

PERSONAL INFORMATION

Date of birth	15 September 1998
Nationality	Italian
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PROFESSIONAL EXPERIENCE

October 2025 – now	Postdoctoral researcher, <i>Photonics and Electromagnetics group, Department of Information Engineering, University of Padova, Italy</i> I study electromagnetic wave propagation in nonlinear and dispersive media, and more generally the classical electrodynamics of continuous media, motivated by applications but approached from a fundamental perspective. My current subjects are slow light and spatiotemporal solitons. A second line of research is the stochastic modelling of the Kerr-mediated interference noise that limits the capacity of high-speed fibre telecommunications links, for which I developed novel analytical approaches inspired by wave turbulence theory, yielding the scaling rules that govern the noise and its statistical properties, and novel efficient estimation algorithms.
October 2022 – September 2025	Doctoral researcher, <i>Quantum Information and Matter group, Department of Physics and Astronomy, University of Padova, Italy</i> My doctorate dealt with nonlinear waves in ultracold quantum gases and in optical media. I developed analytical techniques for dealing with low-energy atomic scattering and the effective interactions for many-body theory. As a peculiar theme connecting optics with ultracold gases I studied the commonalities and differences between matter-wave and optical solitons, including their stability properties. I also studied the effect of fluctuations beyond the mean field in bosonic gases and Josephson junctions. The approach was analytical and numerical, and part of the results were directly compared with a cold-atom experiment in a collaboration.
January 2022 – October 2022	Research collaborator, <i>Photonics and Electromagnetics group, Department of Information Engineering, University of Padova, Italy</i> Before the doctorate I worked on the theory and modelling of nonlinear light propagation in optical fibre. I used a perturbative approach for modelling the impact on realistic telecommunications systems, and explored the same approach for quantum key distribution channels carried alongside classical traffic.

EDUCATION

March 2026	Ph.D. in Physics, <i>University of Padova, Italy</i> Supervisor: Prof. Luca Salasnich. Thesis title: <i>Nonlinear physics of ultracold quantum gases and optical fields in media</i>, defended <i>cum laude</i>.
September 2022	M.Sc. in ICT (photonic engineering track), <i>University of Padova, Italy</i> Supervisor: Prof. Marco Santagiustina. Thesis title: <i>Nonlinear noise in WDM systems: study of classical and quantum channel interaction and capacity</i>, final grade 110/110 <i>cum laude</i>. Weighted exam average: 30/30.
September 2020	B.Sc. in Information Engineering, <i>University of Padova, Italy</i> Thesis title: <i>Dispersione cromatica e non linearità in fibra ottica, un'analisi simulativa</i>, final grade 110/110 <i>cum laude</i>.

RESEARCH VISITS

October 2023 –
February 2024

Barcelona Quantum Monte Carlo group, Universitat Politècnica de Catalunya, Barcelona, Spain

Five months as a visiting researcher with Prof. Pietro Massignan, working on scattering theory and effective interactions for ultracold atoms, and on quantum Monte Carlo methods in collaboration with Prof. Grigori Astrakharchik.

AWARDS AND FUNDING

2022–2025

Full three-year Ph.D. scholarship, University of Padova, $\approx 52,000\text{€}$ gross in total.

2021–2026

Teaching assistantship and tutoring contracts for undergraduate physics and mathematics courses, University of Padova, $\approx 3,600\text{€}$ gross in total.

2021

“Premio di studio del Rotary Club di Padova”, named after Prof. Carlo Giacomo Someda, awarded for academic excellence and service to the community, $\approx 1,200\text{€}$.

PUBLICATIONS

Preprints:

A. Bardin, **F. Lorenzi** and L. Salasnich, *Bosonic Josephson junction dynamics: interplay between quantum and thermal fluctuations*, [arXiv:2604.27809](#) (2026).

Journal articles:

12. **F. Lorenzi** and L. Salasnich, *Dimensional reduction for optical beams with thermal nonlocal nonlinearity*, Front. Phys. **21**, 102203 (2026), in the special issue *Topological Dynamics in Ultracold Atoms and Optics*.
11. **F. Lorenzi**, G. Marcon, L. Schenato, A. Mecozzi, C. Antonelli and M. Santagiustina, *Fast estimation of nonlinear interference noise in distributed Raman-amplified few-mode fiber links*, J. Lightwave Technol. **44**, 4476–4490 (2026).
10. **F. Lorenzi**, M.G. Pelizzo and L. Salasnich, *Variational treatment of the electromagnetic field in an optical cavity with a moving mirror*, J. Phys. Photonics **8**, 015023 (2026).
9. R. Cruickshank, **F. Lorenzi**, A. La Rooij, E.F. Kerr, T. Hilker, S. Kuhr, L. Salasnich and E. Haller, *Experimental observation of single- and multisite matter-wave solitons in an optical accordion lattice*, Phys. Rev. Lett. **135**, 263404 (2025). Selected as Editors’ Suggestion, and featured in APS *Physics*: P.G. Kevrekidis, *An accordion lattice playing a soliton tune*, *Physics* **19**, 3.
8. **F. Lorenzi**, L. Salasnich and M.G. Pelizzo, *Optical cavity in the relativistic regime for laser propulsion*, Phys. Rev. Applied **24**, 034033 (2025).
7. **F. Lorenzi** and L. Salasnich, *Low-energy atomic scattering: s-wave relation between the interaction potential and the phase shift*, Ann. Phys. **537**, e00227 (2025).
6. **F. Lorenzi** and L. Salasnich, *Variational approach to multimode nonlinear optical fibers*, Nanophotonics **14**, 805–813 (2025).
5. L. Salasnich, M.G. Pelizzo and **F. Lorenzi**, *Only-phase Popov action: thermodynamic derivation and superconducting electrodynamics*, J. Phys. A: Math. Theor. **57**, 355302 (2024).
4. **F. Lorenzi** and L. Salasnich, *Atomic soliton transmission and induced collapse in scattering from a narrow barrier*, Sci. Rep. **14**, 4665 (2024).
3. A. Bardin, **F. Lorenzi** and L. Salasnich, *Quantum fluctuations in atomic Josephson junctions: the role of dimensionality*, New J. Phys. **26**, 013021 (2024).
2. **F. Lorenzi**, G. Marcon, A. Galtarossa, L. Palmieri, A. Mecozzi, C. Antonelli and M. Santagiustina, *Nonlinear interference noise in Raman-amplified WDM systems*, J. Lightwave Technol. **41**, 6465–6473 (2023).

1. **F. Lorenzi**, A. Bardin and L. Salasnich, *On-shell approximation for the s-wave scattering theory*, Phys. Rev. A **107**, 033325 (2023).

Proceedings:

F. Lorenzi, L. Schenato, A. Mecozzi, C. Antonelli and M. Santagiustina, *Time-domain two-pulse collisions nonlinear interference noise model for ultra-wideband links*, accepted for ECOC 2026, Málaga, Spain, 20–24 September 2026.

F. Lorenzi and L. Salasnich, *Dimensional reduction for paraxial beams with thermally mediated nonlocal nonlinearity*, in the proceedings of RiNEM 2026, L’Aquila, Italy, 6–9 September 2026.

F. Lorenzi and M. Santagiustina, *Beer–Lambert absorption enhancement in structural and material slow-light*, in the proceedings of RiNEM 2026, L’Aquila, Italy, 6–9 September 2026.

F. Lorenzi, G. Marcon, A. Galtarossa, L. Palmieri, A. Mecozzi, C. Antonelli and M. Santagiustina, *Model for nonlinear interference noise in Raman-amplified WDM systems*, in the proceedings of the **ECOC 2022**, paper Tu1D.4.

Book:

L. Salasnich and **F. Lorenzi**, *Modern Physics: An Invitation to Statistical and Quantum Mechanics*, UNITEXT for Physics, Springer Nature Switzerland, 2nd edition (2025). Italian edition: *Introduzione alla fisica moderna*, **Springer** (2026).

TALKS AND POSTERS

September 2026	“ <i>Dimensional reduction for paraxial beams with thermally mediated nonlocal nonlinearity</i> ” and “ <i>Beer–Lambert absorption enhancement in structural and material slow-light</i> ”. Talks at RiNEM 2026, L’Aquila, Italy.
June 2026	“ <i>Fast estimation of nonlinear interference noise in distributed Raman-amplified few-mode fiber links</i> ”. Talk at ICOP 2026, L’Aquila, Italy.
January 2026	“ <i>Experimental observation of single- and multisite matter-wave solitons in an optical accordion lattice</i> ”. Poster at the QSnow–QTech workshop, Asiago, Italy.
August 2025	“ <i>Variational approach to multimode nonlinear optical fibers</i> ”. Poster and flash presentation at the Bad Honnef Physics School on Ultracold Quantum Matter, Bad Honnef, Germany.
May 2024	Poster on quantum fluctuations in atomic Josephson junctions, co-presented with A. Bardin, at the Atomtronics conference, Benasque, Spain.
January 2024	“ <i>Scattering of matter-wave soliton from a narrow barrier: transmission coefficient and induced collapse</i> ”. Poster at the CAPS Cold Atom Workshop, Barcelona, Spain.
September 2023	“ <i>Universality and effective range corrections in quantum gases</i> ”. Talk at the CMD30/FisMat 2023 conference, Milan, Italy.
September 2022	“ <i>Model for nonlinear interference noise in Raman-amplified WDM systems</i> ”. Talk at ECOC 2022, Basel, Switzerland.

A collection of presentations, seminars and posters is available at lorenzifrancesco.github.io.

PEER REVIEW ACTIVITY

I have reviewed manuscripts for *Nature Communications*, *Physics Letters A*, *Journal of Low Temperature Physics*, *Journal of Lightwave Technology* and *Scientific Reports*.

TEACHING EXPERIENCE

2021–2026	Teaching assistant for the courses “Fisica 1”, “Fisica 2” and “Fondamenti di algebra lineare e geometria”, University of Padova. Supplementary teaching in the physics laboratories of the engineering degrees.
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2021–2024	Tutor for the university inclusion programme, for the courses “Analisi 1” and “Consapevolezza digitale”, University of Padova. Two tutoring assignments in support of the degree courses of the School of Engineering.
February 2021	Information technology teacher, Istituto Tecnico Tecnologico Economico (ITTE) G. Galilei, Arzignano (VI), Italy.

OTHER ACTIVITIES

2025–now	Co-organiser of the FNIP conference (<i>Focus on Optical NeuroImaging and Photonics</i>), University of Padova: responsible for theme T3, <i>simulations and numerical methods</i> , and member of the organising committee for FNIP day 2026. Main organiser of the 9th FNIP webinar, T3 webinar “Open-source tools for classical and quantum optics” (October 2025).
2025	Co-organiser of PRAPI (<i>Ph.D. Research Activity Presentation Initiative</i>), a seminar series in which doctoral students present their research in accessible terms to an audience from other fields of physics, Department of Physics and Astronomy, University of Padova, Italy.
2024–2025	Representative of the Ph.D. students in the Consiglio direttivo of the doctoral school, Department of Physics and Astronomy, University of Padova, Italy.
July 2022	Designer and teacher of the digital literacy course “Strumenti digitali per la cittadinanza” for the general public, Biblioteca comunale G. Bedeschi, Arzignano (VI), Italy. The course was accompanied by a study on the digital divide based on data from the participants.

LANGUAGES

Italian	Native.
English	Fluent, working language for research and teaching. Certified Cambridge FCE B2 in February 2016.
German, Spanish and French	Reading comprehension.

COMPUTER AND TECHNICAL SKILLS

Advanced	Python , Julia , MATLAB , Git/GitHub , L^AT_EX , and running simulations on SLURM -managed clusters. Linux is my operating system of choice, and I am confident with computer maintenance, both software and hardware.
Intermediate	C , C++ , Rust , CST-FEM .
Basic	Fortran , x86 assembly, embedded programming on STM32 (ARM) microcontrollers, SQL .
Electronics	Amateur radio (callsign IU3UHF) and a home radiofrequency electronics lab: qsl.net/iu3uhf .

updated September 2026.